

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	AIML-A4
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: **Secure IoT Intrusion Detection System**

Abstract

The rapid growth of Internet of Things (IoT) devices has significantly increased the number and complexity of cyber-attacks, making Network Intrusion Detection Systems (NIDS) essential for protecting organizational networks. However, traditional centralized intrusion detection systems require organizations to share sensitive network data, creating privacy risks and regulatory challenges. As a result, sectors such as healthcare, finance, government, universities, and enterprises often train intrusion detection models independently, leading to reduced detection accuracy and limited ability to identify emerging cyber threats. Recent research also highlights that existing federated intrusion detection approaches lack strong privacy protection, adaptive feature selection, and comprehensive evaluation across multiple datasets.

*This project proposes **FedPriIDS (Privacy-Preserving Federated Learning for Collaborative Network Intrusion Detection in IoT)**, a secure framework that enables multiple organizations to collaboratively train intrusion detection models without sharing their raw network traffic. The system employs Federated Learning, Differential Privacy, secure model aggregation, and hybrid feature selection to ensure data confidentiality while improving intrusion detection performance. Each participating organization trains a local deep learning model using its own data, and only encrypted model parameters are shared with a central server for aggregation. This collaborative learning approach enhances the global model while preserving organizational privacy.*

*The scope of the project is limited to privacy-preserving collaborative intrusion detection for IoT environments and does not include malware removal, firewall implementation, or real-time production deployment. The proposed framework is intended to complement existing intrusion detection systems rather than replace them. Experimental results reported in the base research demonstrate that the proposed approach achieves detection accuracy above **90%**, reduces preprocessing time by approximately **37–43%**, and improves intrusion detection performance by **8–15%** compared to isolated learning while maintaining strong privacy guarantees. These measurable outcomes indicate that the proposed system can provide an effective, secure, and scalable solution for collaborative cybersecurity in IoT environments.*

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References (Category specific)

Base Paper (Main Reference)

1. Mankotia, S., Conte de Leon, D., & Rimal, B. P. (2026).
FedPrIDS: Privacy-Preserving Federated Learning for Collaborative Network Intrusion Detection in IoT.
Journal of Cybersecurity and Privacy, 6(1), 10.
DOI: 10.3390/jcp6010010
<https://doi.org/10.3390/jcp6010010>

Supporting References

2. Janati Idrissi, M., et al. (2023).
Fed-ANIDS: Federated Learning for Anomaly-based Network Intrusion Detection Systems.
Expert Systems with Applications, 234, 121000.
<https://doi.org/10.1016/j.eswa.2023.121000>
3. Amiri-Zarandi, M., Dara, R. A., & Lin, X. (2023).
SIDS: A Federated Learning Approach for Intrusion Detection in IoT Using Social Internet of Things.
Computer Networks, 228, 110005.
<https://doi.org/10.1016/j.comnet.2023.110005>
4. Liu, Y., Kang, Y., Zou, T., Pu, Y., He, Y., & Ye, X. (2024).
Vertical Federated Learning: Concepts, Advances, and Challenges.
IEEE Transactions on Knowledge and Data Engineering, 36, 3615–3634.
<https://doi.org/10.1109/TKDE.2024.3352628>

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